

A Sentiment analysis-based hotel recommendation using TF-IDF Approach

Ram Krishn Mishra
Department of Computer Science
BITS Pilani, Dubai Campus
Dubai, United Arab Emirates
rkmishra@dubai.bits-pilani.ac.in

Siddhaling Urolagin
Department of Computer Science
BITS Pilani, Dubai Campus
Dubai, United Arab Emirates
siddhaling@dubai.bits-pilani.ac.in

Angel Arul Jothi J
Department of Computer Science
BITS Pilani, Dubai Campus
Dubai, United Arab Emirates
angeljothi@dubai.bits-pilani.ac.in

Abstract—Speedy evolution in the internet world has enforced many users to use internet services for daily basis work. Tourism industry has played a vital role for internet users by offering economical and comparable prices for hotel booking. In current scenario it has been observed that many users share their views by giving their feedback in different forms like blogs, web forums, social media etc. In this research we have used term frequency-inverse document frequency (TF-IDF) and Cosine Similarity to provide more options of hotels based on their reviews. TF-IDF helps to find the weight value of terms or documents frequency and the cosine similarity helps to extract similar kinds of values from the sentiment dataset.

Keywords—Recommendation Systems, TF-IDF, Sentiment Analysis.

I. INTRODUCTION

The tourism industry is one of the important sectors in many countries. It contributes significantly to the economy of many countries. In recent decades, there is tremendous growth in tourism. As per the data provided by United Nation World Tourism Organization (UNWTO), international tourist arrivals in the year 2017 to France are 86.9 million and to Spain are 88.8 million with a growth 5.1 % and 8.6% respectively when compared with the previous year. The total international tourist arrivals have been counted 1.323 billion worldwide [1] and contributed 10.4% in the world's GDP. As per the report provided by the World Travel and Tourism Council (WTTC) in the year of 2017 tourism has contributed US\$ 8.3 trillion, which is 10.4% of the world's GDP [2].

Tourism encourages various business, services, and other sectors. In tourism, people travel to and stay in the places outside their usual environment for happiness and business purpose. This industry has three major components. They are attraction, accommodation, and transport. Tourists are using the tourism services in the form of inbound, outbound and domestic. When a tourist visits the places from one province to another province in own country, it is called inbound tourism. In outbound tourism, tourist select the places for visit in another country. Domestic tourism encourages in same city as entertainment, movie etc. Tourism sectors size is measured by different parameter. It covers forestry, hospitality and other industries like local product manufacturing, etc.

During the last few years, information technology is playing an important role in the field of tourism. It has been seen that information technology has improved the quality of operation efficiency and customer satisfaction. It helps in the reduction of cost for reservation, transport, hotels, etc. and working as a local guide by using the Global Positioning System (GPS) Map for the travelers while moving from one place to another place [3]. Chain of hotels are being connected

by a communication system, which are in a different location. It is helping to give new guest requests, housekeeping information, reservation extension, and other facilities by making information easier to the client.

Nowadays machine learning plays a vital role in prediction by learning the experience from the data. It is widely used in the field of data analytics where a complex model is designed for predictive analytics to get some fruitful results for the commercial purpose. It has become important to have a recommendation system which can give a clear picture intelligently. The recommendation system can be developed based on user preferences, existing history of visitors or both to provide user valuable suggestions. The tourism industry has been transformed tremendously in terms of processing and facilitating new things using machine learning techniques [4,5]. Machine learning helps tourists to compare hotel rates, chatting via Chatbots, selecting rooms and analyzing people based on user feedback reviews. The customer feedback reviews contribute an important role in making right decision. Using ensemble methods multiple learners prepared on opinion datasets and merged them into a combined model which is helping to find out the new trends of tourists [6].

Many organizations collect online reviews data for detecting and extracting the opinion. There is huge amount of data being gathered every day in terms of user opinions, reviews, and suggestions. Major challenges in opinion mining is to extract the usual information from unstructured data [7]. The data related to user opinion and review can be analyzed to extract useful information to help tourist find out best destination, location, hotels, in less time. Moreover, the textual information can be further analyzed to understand difficulties, limitations, and drawbacks suggested by tourist. These analyzed data will be useful to improve tourist related services. The multi-source review data can be summarized for the benefit of end users.

The main objective of this research is to find similar kind of hotels based on the reviews submitted by the users. We use the term frequency-inverse document frequency (TF-IDF) and cosine similarity to find the similarities between hotels based on reviews and recommend it to users.

II. BACKGROUND

Mike Thelwall has explained in [8] about the positive and negative options by applying the sentiment analysis in the text dataset and also given the different methods to work. Most of the search engines are providing the destination review to make the proper decision to visit the place. Authors have applied the sentiment classification method by mining the reviews from travel blogs and compared the Naïve Bayes,

Support Vector Machine(SVM) and the character-based N-gram model [9]. C. A. Martín et. al. in [10] have developed several classifiers to determine the sentiment of reviews collected from various platforms of hotel services. The classification techniques compared in this work are based on convolutional neural networks (CNN) and Long Short-Term Memory Networks (LSTM).

Cristian Bucur has proposed the review analysis systems for the hotel by collecting the data from the internet and applied the opinion mining techniques [7]. It has also been observed that results have been improved using text mining in domain-specific measures. Fuzzy aspect-based opinion classification has been developed in [11] which explore the user opinion efficiently with better performance and classification accuracy. Relevant trends have been analyzed on hotel ratings and reviews data. As per overall ratings, the top 10 hotels have been listed through online and backend work been applied through off-line using Linear regression to identify the similar type of rating. Dawei Chen et.al. in [4] suggested a probabilistic model to ensemble the outcome of Point of Interest(POI) ranking and the POI to POI movements.

Flickr photos data set been used for experiments and suggest the tour recommendation approach on maximum likelihood sequences. The combined PoI ranks, and routes allow learning the feature-driven based approach. Decision support systems have been developed by Shah Jahan Miah et. al in [12] to predict the tourist behavioral pattern on specified location using the geotagged photos shared in social media sites like Flickr etc. Call details record been used to predict the comprehensive prediction of tourists in [13] by applying the machine learning and deep learning techniques. In [6] authors have proposed the opinion mining from the review data collected from tripadvisor.com to boost the tourism along with economic growth.

Ajay k. Tripathi et.al in [14] suggested that the Internet of Things(IoT) based solution for tourist mobility and analyzed the challenges and issue of adopting it. It has also been suggested the real-time assistant to a traveler from one place to another place in which the MAP will guide with the best possible solution at a given instant. In [15] the contribution is made on predicting the visiting time. The researchers have used Google Map and Crowd sourcing data and provide valid information to tourist. Several regression techniques are employed such as ordinal logistic regression, multiple linear regression. The developed model used to predict the visiting time of 402 famous destinations. Yi Fei Li and Han Cao have been used LSTM for tourism flow prediction which has the better performance as compared ARIMA model and BPNN [16]. A new algorithm has been proposed to personalized POIs travel route when a traveler is coming to new place. Using this method authors have integrated the multi-source tourism data by generating the POIs knowledge and POIs visit sequences. The suggested systems have been tested on real tourism data for proper validation[17]. Possible research challenge and opportunities have been discussed in [18] and how the information technology is accelerating and acting as a market-driven tools for the tourism.

III. METHODOLOGY

In this work, we propose to build a hotel review analysis and recommendation system as shown in Figure 1. We start with the data collection from the available source followed by data preprocessing and attribute selection. We then use

visualization techniques to analyze the data. Finally, we use TF-IDF and cosine similarity for recommendation.

A. Raw Data Collection

This is the first step in which required data is collected from the available source. One of the most challenging steps is to identify the required data source. In this research, we have used 515K Hotel Reviews Data in Europe available on Kaggle. The dataset contains reviews on hotels present in different cities in Europe. The dataset consists of attributes such as hotel address, additional number of Scoring, review date, average score, hotel name, reviewer nationality, negative review, review total negative, word counts, total number of reviews, positive review, review total positive word counts, total number of reviews reviewer has given, reviewer score, tags, days since review, latitude, longitude.

B. Data Preprocessing

Data preprocessing step validates the fitness of the data. Preprocessing involves data cleaning and making the data in accurate format. From the original dataset we select average score, hotel name, negative review, positive review and review date for further data analysis.

C. Data Analysis

This phase involves analyzing the raw data using visualization techniques in order to understand the trends and patterns in the data. This phase can rely on simple techniques to complex machine learning algorithms.

D. TF-IDF Based Recommendation

TF-IDF stands for “Term Frequency-Inverse Document Frequency”. TF-IDF is used to measure the importance of a word in a given document. TF-IDF is widely used when we are trying to extract some usual information from the text or reviews data. TF-IDF method assigns a weight to a word based on its occurrence in a document. The frequency of the word provides an approach for scoring or ranking of data and finding the similarity relation among the words. This can be formally defined as follows:

TF (Term Frequency): TF measures the frequency of a term (t) in a document. It is given by equation (1)

$$TF(t) = \frac{(\text{Number of times term } t \text{ appears in a document})}{(\text{Total number of terms in the document})} \quad (1)$$

Inverse Document Frequency (IDF): TF gives equal importance to all words but IDF measures how important a word is. IDF can be computed using equation (2) given below

$$IDF(t) = \frac{(\text{Total number of documents})}{(\text{Number of documents with word } t \text{ in it})} \quad (2)$$

Finally, TF-IDF is obtained as the product of term frequency and inverse document frequency and is given by equation (3)

$$TF - IDF = TF * IDF \quad (3)$$

In this research, we compute the TF-IDF for the reviews in the dataset.

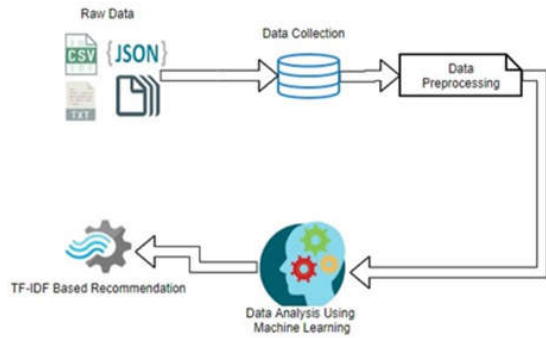


Fig. 1. Hotel review analysis and recommendation system

IV. EXPERIMENT RESULTS

In this work, we used Python3 Libraries to perform the analysis and recommendation of hotel reviews data. This section shows the results of analyzing the dataset and the TF-IDF based recommendation system.

A. Analyzing the Dataset

a) **Average score versus number of hotels:** Figure 2 is a graph plotted between the average rating score and numbers of hotels that come under the selected rating category. As per Figure 2 the average score for most of the hotels are between 7.4 to 9.5.

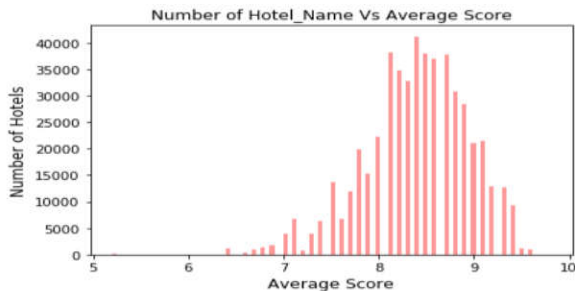


Fig. 2. Plot between number of hotels and average score

b) **Top 10 hotels:** Table 1 shows top 10 hotels based on the review scores given by reviewers.

TABLE I. TABLE 1. REVIEW SCORES OF TOP 10 HOTELS

Hotel Name	Review Score
Ritz Paris	9.72
Hotel Casa Camper	9.72
41	9.71
H tel de La Tamise Esprit de France	9.69
Le Narcisse Blanc Spa	9.68
H10 Casa Mimosa 4 Sup	9.66
Hotel Eiffel Blomet	9.65
Hotel The Serras	9.62
45 Park Lane Dorchester Collection	9.62
The Soho Hotel	9.60

Figure 3 shows the bar chart of top 10 hotels and their review score. It can be noted from Figure 3 that review score of top 10 hotels vary between the 9 and 10.

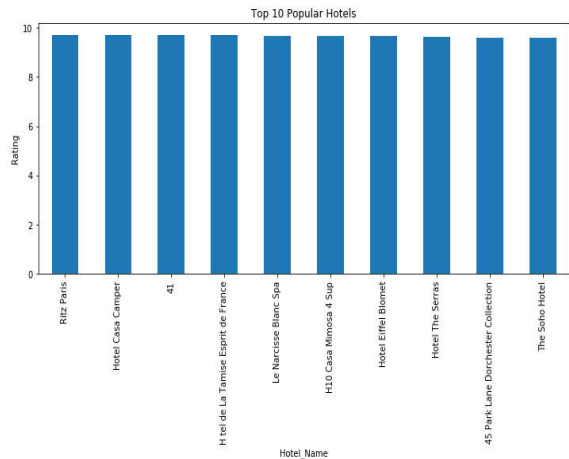


Fig. 3. Bar Chart between top 10 hotels and their rating

c) **Analysis of top 10 hotels on specific date:** We further analyzed the top 10 hotels with respect to reviews on specific date. Figure 4 depicts the trends of average score for the top 10 hotels for some specific date using a graph plot.

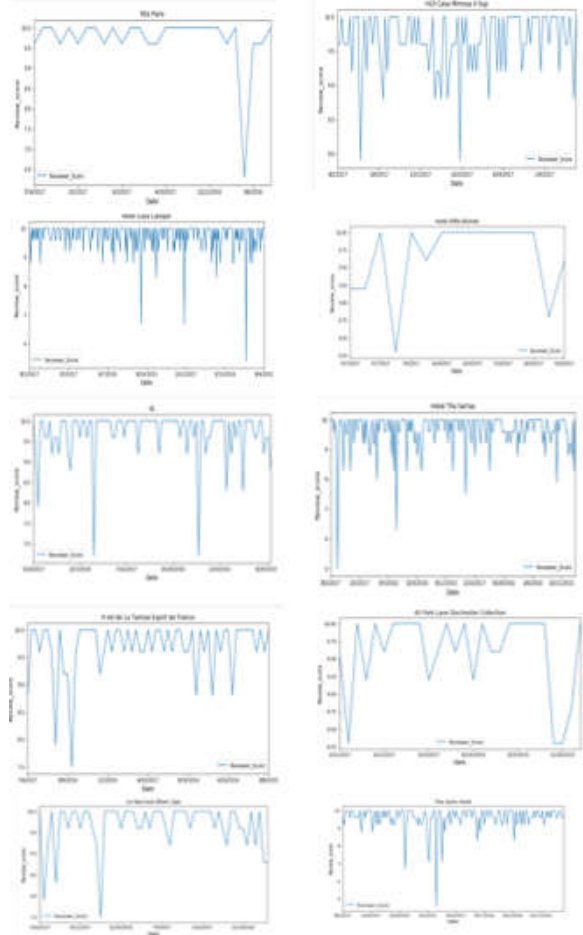


Fig. 4. Figure 4: Graph plot between review score and date (a) Ritz Paris (b) Hotel Casa Camper (c) 41 (d) H tel de La Tamise Esprit de France (e) Le Narcisse Blanc Spa (f) H10 Casa Mimosa 4 Sup (g) Hotel Eiffel Blomet (h) Hotel The Serras (i) 45 Park Lane Dorchester Collection (j) The Soho Hotel

d) **Negative and positive word cloud formation:** We intend to form the negative and positive word cloud for the dataset. For this purpose, we selected the hotel names and the review texts given by the reviewers from different locations. Figure 5 shows the negative and positive reviews word obtained by using the Python NLTK library.



Fig. 5. (a) Negative Words Cloud (b) Positive Words Cloud

B. TF-IDF Based Recommendation

Python sklearn provide the TfidfVectorizer library which process any document in three steps, firstly it computes the word counts, secondly it finds the IDF values and finally find the scores of terms in the dataset.

We supplied the hotel reviews dataset as input to the TfidfVectorizer. As per the experiment there are 1493 hotels and more than 81000 different words which have been used in reviews given by the reviewers. Table 2 is the output of document id and their weight values from 0th index to N-1 index values.

TABLE II. WEIGHT VALUES FOR EACH WORD ASSIGNED BY TF-IDF

Term	TF-IDF Weight Values
(0,69268)	0.014599755728994
(0,66435)	0.007329917999188
(0,70675)	0.015857483517701
:	:
(1491,10503)	0.128359563950412
(1491,56494)	0.008187795861974
(1491, 22271)	0.021321839690902
(1491, 31291)	0.056847389819470
(1491 60772)	0.040385612808382
(1491,12659)	0.106252239852796

The TF-IDF converts the terms or words to weight values and eliminates most of the common words during the processing time. Thus after the application of TF-IDF to the dataset a review is converted into a vector of weights of the terms in the review.

Similarity measure for Recommendation:

Cosine similarity is then used to measure the similarity between two reviews. Cosine similarity measures the cosine of the angle between two vectors in a multidimensional space. Let x and y represent two vectors (reviews), n denote the total number of components of the vectors and each component of the vector x and y be denoted as x_i and y_i . Then, cosine similarity is given by equation (4).

$$\text{Cosine Similarity} = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}} \dots \dots \dots (4)$$

Similar kind of hotels can be found by giving a hotel name as an input and computing the cosine similarity. Table 3 is the result of 10 similar kind of hotels that is recommended based on input hotel name "Hotel Pulitzer".

Table 3: Results of similar recommended hotels for input hotel name "Hotel Pulitzer"

Index Value	Hotels Name
632	Royal Passeig de Gracia
1218	Hotel Espa a Ramblas
96	Avenida Palace
490	H10 Urquinaona Plaza
487	H10 Metropolitan 4 Sup
483	H10 Cubik 4 Sup
212	Catalonia Plaza Catalunya
1068	Negresco Princess 4 Sup
580	Hotel Barcelona Catedral
211	Catalonia Passeig de Gr cia 4 Sup

V. CONCLUSION AND FUTURE SCOPE

It has been seen that sometimes it is difficult to find proper hotels on numerical rating. Hence, sentiments reviews of customers help organizations to suggest proper hotels to the customers. In this research, we used TF-IDF and cosine similarity to recommend similar type of hotels based on the reviews of different users from the globe. This research can be extended with more efficient algorithms to design more accurate recommendation systems. Recommendation of hotels for the selected destination may vary as per different time periods, so more studies can be extended using time series data by understanding the sentiments of the reviewers

REFERENCES

- [1] UNWTO, "2018 Edition UNWTO International Tourism Trends 2017," pp. 1–20, 2018.
- [2] WTTC, "The Comparative Economic Impact of Travel & Tourism," Wttc, p. 40, 2012.
- [3] E. Šimková, "The role of information technology in the development of rural tourism and its presentation," 2013 12th Int. Conf. Inf. Technol. Based High. Educ. Training, ITHET 2013, pp. 1–6, 2013.
- [4] D. Chen, C. S. Ong, and L. Xie, "Learning Points and Routes to Recommend Trajectories," pp. 2227–2232, 2016.
- [5] O. Claveria and S. Torra, "The appraisal of machine learning techniques for tourism demand forecasting," no. March, pp. 59–89, 2017.
- [6] N. Kumar, C. Puri, Kush, Akhil, and Gaurav, "Opinion Ensembling for Improving Economic Growth through Tourism," Procedia Comput. Sci., vol. 122, pp. 237–244, 2017.
- [7] C. Bucur, "Using Opinion Mining Techniques in Tourism," Procedia Econ. Financ., vol. 23, no. October 2014, pp. 1666–1673, 2015.
- [8] M. Thelwall, Big Data and Innovation in Tourism, Travel, and Hospitality. Springer Singapore, 2019.
- [9] Q. Ye, Z. Zhang, and R. Law, "Sentiment classification of online reviews to travel destinations by supervised machine learning approaches," Expert Syst. Appl., vol. 36, no. 3 PART 2, pp. 6527–6535, 2009.
- [10] C. A. Martin, J. M. Torres, R. M. Aguilar, and S. Diaz, "Using Deep Learning to Predict Sentiments: Case Study in Tourism," Complexity, vol. 2018, pp. 1–9, 2018.
- [11] M. Afzaal, M. Usman, A. C. M. Fong, S. Fong, and Y. Zhuang, "Fuzzy Aspect Based Opinion Classification System for Mining Tourist Reviews," Adv. Fuzzy Syst., vol. 2016, 2016.
- [12] "28_Ref_A Big Data Analytics Method for Tourist Behaviour Analysis."
- [13] N. C. Chen, W. Xie, R. E. Welsch, K. Larson, and J. Xie, "Comprehensive Predictions of Tourists' Next Visit Location Based on Call Detail Records Using Machine Learning and Deep Learning Methods," Proc. - 2017 IEEE 6th Int. Congr. Big Data, BigData Congr. 2017, pp. 1–6, 2017.

- [14] A. K. Tripathy, P. K. Tripathy, N. K. Ray, and S. P. Mohanty, "ITour: The Future of Smart Tourism: An IoT Framework for the Independent Mobility of Tourists in Smart Cities," *IEEE Consum. Electron. Mag.*, vol. 7, no. 3, pp. 32–37, 2018.
- [15] I. Hapsari, I. Surjandari, and Komarudin, "Visiting time prediction using machine learning regression algorithm," *2018 6th Int. Conf. Inf. Commun. Technol. ICoICT 2018*, vol. 0, no. c, pp. 495–500, 2018.
- [16] Y. Li and H. Cao, "Prediction for Tourism Flow based on LSTM Neural Network," *Procedia Comput. Sci.*, vol. 129, pp. 277–283, 2018.
- [17] Bin C., Gu T., Sun Y., Chang L., Sun W., Sun L. (2018) "Personalized POIs Travel Route Recommendation System Based on Tourism Big Data. In: Geng X., Kang BH. (eds) *PRICAI 2018: Trends in Artificial Intelligence. PRICAI 2018. Lecture Notes in Computer Scien.*" vol. 11013, p. 11013, 2018.
- [18] Z. Xiang, "From digitization to the age of acceleration: On information technology and tourism," *Tourism Management Perspectives*, vol. 25, pp. 147–150, 2018.